

Ali Parchekani

PhD-Level STEM Tutoring & Academic Mentorship

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Profile

University of Toronto Electrical and Computer Engineering PhD Candidate offering private, high-touch STEM tutoring and academic mentorship for ambitious high-school, pre-university, undergraduate, graduate, and professional learners. I help students build real understanding in technical subjects through step-by-step explanation, guided problem solving, exam-focused preparation, and project-based learning.

My background spans 40+ distinct University of Toronto courses (100+ TA/course-staff appointments) across mathematics, statistics, programming, computer science, physics, AI, data science, networks, and engineering fundamentals, plus instructor roles in University of Toronto pre-university statistics, AI, and game-design courses. Student evaluations consistently single out clarity, patience, and individualized support.

What Students Say

"My tutor went over the specific questions that related to our quizzes; the way of explaining is straightforward and helpful." — first-year calculus student

"The steps were broken down and explained thoroughly — there was clear reasoning behind each one." — calculus student

"Very understanding and kind, with great communication skills; explains things in a way that's easy to follow." — calculus student

"I felt I learned more from asking questions than from following the screen — I'm glad I had him as the instructor." — Games, Strategy & AI student

In a recent pre-university course taught as instructor of record, every responding student rated the course "Excellent" or "Very good," and all would recommend it. (Full anonymized evaluations available on request.)

Best-Fit Students

- High-school, IB/AP, and pre-university students preparing for STEM-heavy university programs.
- First-year university and college students in calculus, physics, statistics, programming, or engineering.
- Undergraduates needing course rescue, exam preparation, assignment support, or project mentoring.
- Graduate students and professionals building Python, MATLAB, AI, data-science, or machine-learning foundations.
- Motivated students who benefit from structured plans, accountability, and clear progress communication.

Core Subjects

High-demand foundations: calculus, advanced functions, linear algebra, probability, statistics, physics, mechanics, programming fundamentals, Python, MATLAB.

University-level STEM: data structures, algorithms, data science, AI foundations, machine learning, signals and systems, engineering mathematics, computer networks.

Advanced mentorship: AI/ML projects, data-science projects, research preparation, technical writing, simulation workflows, academic planning.

Teaching Approach

1. **Diagnostic first session:** identify gaps, goals, course constraints, and timeline.
2. **Step-by-step concept repair:** rebuild foundations before moving to difficult problems.

3. **Guided practice:** work through examples, then gradually shift responsibility to the student.
4. **Exam strategy:** convert course content into review plans, practice sets, and test-taking routines.
5. **Project support:** help students plan, debug, document, and present technical work.
6. **Progress communication:** optional parent/student updates, next steps, and study priorities after sessions.

Selected Credentials

- PhD Candidate, Electrical and Computer Engineering, University of Toronto — expected August 2026.
- Instructor for University of Toronto pre-university courses in statistics, critical thinking, AI, and game design.
- Recipient of the University of Toronto Doctoral Completion Award (two years) and a multi-year University of Tehran undergraduate scholarship.
- Supported instruction across 40+ distinct UofT courses spanning calculus, linear algebra, probability/statistics, Bayesian statistics, stochastic processes, programming, data structures, AI, ML, data science, networks, physics, and signals/systems.
- Published IEEE research in wireless sensing, radar signal processing, optimization, and applied machine learning.

Project-Based Learning

I designed and taught a 15-session Game Studies and AI course in which students aged 15–18 built playable game prototypes in GDevelop while learning game mechanics, debugging, strategy, and introductory AI ideas. A private student-project portfolio is available on request where consent allows.

Availability

Available for Toronto/GTA in-person tutoring and online tutoring for Canadian and international students. Rates are provided by consultation based on level, subject, format, and support needs.